MICHEAL T. ADEDIRAN

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Seeking a mandatory industrial training (SIWES) from April 2027 – September 2027

AREAS OF INTEREST

- Vehicle Assembly – Production Line, Fitment and Sub-Assembly Build
- Automotive Systems – Chassis, Drivetrain and How the Parts Work Together

EDUCATION

BSc Mechanical Engineering, University of Lagos, Nigeria Expected 2028
CGPA 4.06/5.00, Second Class Upper (2:1)

TECHNICAL SKILLS

CAD: SolidWorks (CSWA certified; Weldments, Sheet Metal), Creo (surfacing), Blender (rendering)
Simulation: ANSYS Mechanical (static structural, fatigue), ANSYS Fluent (2D RANS), SolidWorks Simulation, MSC Adams
Programming and software: Python, MATLAB, Microsoft Office (Excel, Word, PowerPoint, Access, Outlook)

INDEPENDENT ENGINEERING PROJECTS

198-Part Motorcycle: Design, FEA and Manufacturing Study 2026 | github.com/brickflows/motorcycle

- Modeled and assembled a 198-part twin-cylinder motorcycle in SolidWorks (Weldments frame, engine, 5-speed gearbox, suspension); surfaced the bodywork and fuel tank in Creo Parametric.
- Redesigned the motorcycle frame in ANSYS Mechanical, reducing peak stress by 76% and improving safety margins and structural integrity.
- Swingarm and output shaft: hand calculations within 3% of ANSYS reactions; shaft fatigue FoS 1.74 (Goodman). Rejected a mesh-singularity peak and showed by Lewis and AGMA checks that the first-gear teeth fail.
- Wrote manufacturing plans and dimensioned drawings for the carburized 8620 steel output shaft and the cast A356-T6 aluminum engine case.

Transidime Pocket Translator: Enclosure Design — 3rd place and grant, Innovate Local 2025

- Designed the 3D-printed enclosure around a Raspberry Pi Zero and ReSpeaker HAT; after the first print cracked, increased the wall thickness from 1 mm to 2.5 mm and reprinted.

Front Wing in Ground Effect: CFD Study 2026 | github.com/brickflows/front-wing-ground-effect

- Swept an inverted NACA 4412 through 11 ride heights in ANSYS Fluent (2D RANS, k-ω SST, moving ground, wall y+ ≤ 1.8), after checking the h/c 1.0 case against reference lift and drag.
- Downforce rose 47% as gap closed before stalling at h/c < 0.20; traced stall to 57% chord separation on the suction surface and recommended a ride-height window of 0.20–0.25 h/c for stable aerodynamic performance.

EXPERIENCE

Research Volunteer, Decentralized Grid Task Force, University of Lagos 2026
Title: Institutional Design for Circular Battery Systems in Electrified Transport Transitions: A Global South Implementation Architecture. Supervisor: Assoc. Prof. Modupe A. Onitiri. In coordination with Fraunhofer ISI, Germany.

- Mapping the operational and planned storage capacity of solar mini-grids and commercial microgrids across Nigeria, and collecting sizing and inverter-interface data to assess second-life EV batteries for grid storage.

Mechanic Apprentice, D'leye Automobile: 2022

- Diagnosed and repaired automotive systems over six months, including full engine teardown and rebuild — hands-on exposure to component wear, tolerances, and assembly sequences that inform design-for-manufacturing decisions.

TEAM PROJECTS AND AWARDS

- **NAVI navigation cane for blind users** (team of 4): regional stage, AAKRUTI Innovation Competition 2026 (Dassault Systèmes). Ran structural simulations of the telescoping shaft in SolidWorks Simulation and produced the renders.
- **Transidime plug-in translation hub** (team): 3rd place and a grant, Innovate Local 2025. Designed the 3D-printed enclosure; after the first print cracked, thickened the wall from 1 mm to 2.5 mm and added 0.5 mm cover clearance.
- **HealthMate AI medical assistant** (team lead): 2nd place, AI Campus Network Hackathon 2025, Center for Journalism Innovation and Development (CJID). *Press coverage*: The Nation.
- **Ogun State bursary award** (₦50000) for undergraduate study.

LEADERSHIP AND TEACHING

- Public Relations Officer (2026) and Campus Ambassador (2025), Lagos SolidWorks User Group: organized and presented at community events.
- SolidWorks tutor, University of Lagos Faculty of Engineering (2026): ran a tutorial series for coursework and exams.
- Participant, One Million Leaders Africa (OMLA) leadership and social-innovation fellowship training (2026).